FAYÇAL DRISSI

POST-DOCTORAL RESEARCHER · UNIVERSITY OF OXFORD · FINANCIAL TECHNOLOGY

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EDUCATION

2024-2026 **University of Oxford**, Postdoctoral Research.
2020-2023 **Université Paris 1 Panthéon-Sorbonne**, PhD in Applied Mathematics.
2013-2014 **Université Paris Dauphine-PSL**, MSc in Financial Engineering
2009-2012 **IAE Graduate School of Management & ENSIMAG**, *Diplôme ingénieur* Applied Mathematics & Quant Finance
2008-2009 **Classes Préparatoires**, Advanced Courses in Mathematics and Physics. Distinction.

RESEARCH

REVISE AND RESUBMIT

1. Equilibrium Liquidity and Risk Offsetting in Decentralised Markets (with S. Jaimungal and X. Wu) (2025). *Operations Research.* link.

PUBLICATIONS

1. Strategic Bonding Curves in Automated Market Makers (with Á. Cartea, L. Sánchez-Betancourt, D. Siska, and L. Szpruch) (2026).
   Mathematics of Operations Research. link.
2. Multi-Task Gaussian Process Bandits for Regime-Switching in Optimal Execution (with Á. Cartea and G. Palmari) (2026).
   SIAM Journal on Financial Mathematics. link.
3. Decentralised Finance and Automated Market Making: Execution and Speculation (with Á. Cartea and M. Monga) (2025).
   Journal of Economic Dynamics and Control, Volume 177, 105134. link.
   Presented at: WBS Gillmore Centre Annual Conference on DeFi & Digital Currencies, Warwick Business School, 2023.
4. Decentralised Finance and Automated Market Making: Predictable Loss and Optimal Liquidity Provision (with Á. Cartea and M. Monga) (2024).
   SIAM Journal on Financial Mathematics, 15(3), 931-959. link.
5. Multi-asset Optimal Execution and Statistical Arbitrage Strategies under Ornstein-Uhlenbeck Dynamics (with P. Bergault and O. Guéant) (2022).
   SIAM Journal on Financial Mathematics, 13 (1), 353-390. link.

WORKING PAPERS

1. Traders on GPT: AI and Cognitive Capital in Financial Markets (with A. Barbon) (2026). link.
2. The Viability of Blockchain Markets under Discrete Clearing and Paid Priority (with A. Capponi and Á. Cartea) (2025). link.
   Presented at: NBER Summer Institute 2026 (Cambridge, MA)
   6th Annual CBER Conference (New York, 2026)
   EEA-ESEM (Econometric Society), Dublin, 2026
   The Microstructure Exchange seminar series.
3. Liquid Staking and the Limits of Policy (with Z. Feinstein and B. Williams) (2026). link.
   Presented at: CBER Crafting the Cryptoeconomy Conference (Columbia University, 2025)
   Designing DeFi Conference (Columbia Business School, 2026).
4. The Economics of Regulating Tokenized Securities (with Á. Cartea and F. Saleh) (2026). link.
   Presented at: ToDeFi – Torino Decentralized Finance Conference, 2026 (keynote).
5. Adaptive Lookback for Bayesian Decision Making (with G. Duran-Martin, G. Palmari, and Á. Cartea) (2025).
6. Agentic AI in Proof-of-Stake Blockchains (with Á. Cartea, M. Egri, and P. Chang).
7. Bitcoin As A Hedge Asset (with K. John and F. Saleh).

Conference and other publications

1. DARE: The Deep Adaptive Regulator for Control of Uncertain Continuous-Time Systems (with H. Waldon, Y. Limmer, U. Berdica, J.N. Foerster, and Á. Cartea) (2024).
   ICML 2024 – Foundations of Reinforcement Learning and Control. link.
2. Execution and Statistical Arbitrage with Signals in Multiple Automated Market Makers (with Á. Cartea and M. Monga) (2023).
   IEEE 43rd International Conference on Distributed Computing Systems, 37-42. link.
3. Predictable Losses of Liquidity Provision in Constant Function Markets and Concentrated Liquidity Markets (with Á. Cartea and M. Monga) (2023).
   Applied Mathematical Finance, 30(2), 69-93. link.
4. Solvability of Differential Riccati Equations and Applications to Algorithmic Trading with Signals (2022).
   Applied Mathematical Finance, 29 (6), 457–493. link.

TEACHING and SUPERVISION[1]

Lecturer

2025-2026 **University of Oxford, Saïd Business School**, MSc Financial Economics (MFE).
Lecturer, Continuous-time Finance, 24 hours. link
2024 **University of Oxford, Maths Institute**, MSc Mathematical and Computational Finance.
Lecturer, Microstructure and Algorithmic Trading, 8 hours. link
2022 **Université Paris 1 Panthéon-Sorbonne**, MSc modélisation aléatoire finance & data science.
Lecturer, Computer C++ and Applications to Finance, 21 hours. link

PhD co-supervision

2025 **Gabriel García-Arenas**, University of Oxford (with Prof. Álvaro Cartea)
2025 **Egri Máté**, University of Oxford (with Prof. Álvaro Cartea)
2025 **Gianluca Palmari**, Scuola Normale Superiore (with Prof. Fabrizio Lillo)

MSc thesis supervision

2023-2024 **University of Oxford, Mathematical Institute**, 6 students (DeFi and deep learning/control)
2023 **University of Oxford, Department of Statistics**, 2 students (blockchain data analysis)
2021 **ENSAE ParisTech**, 3 students (Almgren-Chriss model)

SCHOLARSHIPS & GRANTS

Research Grants

2025 **Uniswap Foundation**, Research grant, £40 000.
2023 **Chaire Fintech at University Paris Dauphine - PSL**, Research Grant, €10 000.

Scholarships and Awards

2023 **Best PhD Thesis prize**, EURO Working Group for Commodities and Financial Modelling.
2023 **PhD Prize in Maths and Data Science**, G-Research.
2022-2023 **PhD Bursary**, Oxford-Man Institute of Quantitative Finance.
2020-2022 **PhD Bursary**, New York Life Insurance.
2009-2012 **Excellence Scholarship**, Moroccan Ministry of Higher Education, Scientific Research and Innovation.

[1] Details and lecture notes: www.faycaldrissi.com/teaching/

ACADEMIC ACTIVITIES

Guest Editor	**Special Issue on Decentralized Finance** International Journal of Theoretical and Applied Finance. link
Topic Editor	**Journal of FinTech** World Scientific.
Referee	Operations Research, Management Science, Annals of Operations Research, Mathematical Finance, Finance & Stochastics, SIAM, Applied Mathematical Finance, International Journal of Theoretical and Applied Finance, Digital Finance, Quantitative Finance, Ledger, Frontiers of Mathematical Finance.
Curator	**Quantitative Finance Research Newsletter (OMIReNew)** Oxford-Man Institute.

INDUSTRY EXPERIENCE

2017-2021	**New York Life Investment – Candriam**, Quantitative trading, Systematic & Global Macro.
2016-2017	**AXA Investment Managers**, Quantitative research, rates derivatives & Systematic investment.

CONFERENCES / SEMINARS

Organized

2026. OMI & Saïd Business School Conference on Fintech and Blockchain Economics. **link**.
2025. INFORMS Annual Meeting 2025 session on Blockchain Economics and Decentralized Finance. **link**.
2025. SIAM Minisymposium on Decentralised Finance and Market Microstructure. **link**.
2025. The Fields Institute Workshop on Decentralized Finance and Market Microstructure. **link**.
2025. Oxford-Harvard Conference on Decentralised Finance and Market Microstructure. **link**.
2023-Present. OMI Finance Seminar monthly series, OMI Students Seminar, and OMI Crossroads Seminar. **link**.
2023-2024. DeFox weekly seminar series on decentralised finance, University of Oxford. **link**.
2023. Financial Technology Conference, University of Oxford. **link**.
2023. Blockchain@X-OMI Workshop on Blockchain and Decentralized Finance, Ecole Polytechnique/Oxford. **link**.

Invited

2025. CBER Symposium – Rising Stars of Cryptoeconomics. **link**.
2025. CBER Conference. Columbia University's Center for Digital Finance and Technologies. **link**.
2025. AI & Digital Assets (AIDA) Workshop. University of Edinburgh, Business School. **link**.
2025. SIAM-BFS Mathematical Finance Online Seminar. **link**.
2025. Crypto Academic Camp. Edge Esmeralda. Uniswap Foundation.
2024. Finance & Stochastics seminar, Imperial College London .
2024. Stochastic Finance @Warwick seminar, Warwick University.
2024. AMMs beyond constant functions. *Frontiers in DeFi. Complexity Science Hub, Vienna.* **link**.
2024. Bonding curves of AMMs. *Stevanovich Center for Financial Mathematics, University of Chicago.* **link**.
2024. Equilibrium liquidity in AMMs. *Seminar on DeFi, Harvard University*.
2024. Price impact in Quote Driven Markets. *Seminar on Microstructure, Imperial College London*. **link**.
2024. OTC Market Making with Competition and Impact. *Online Finance Seminar, University of California, Los Angeles*.
2023. AMM design and predictable losses. *Mathematical Institute Seminar, University of Oxford*.
2023. Liquidity provision in AMMs. *QuantMinds International Deep Dive & Digital Assets*.
2023. Optimal liquidity provision in AMMs. *BlockSem seminar, Ecole Polytechnique*.
2023. Bandits for algorithmic trading. *Machine Learning Conference, University of Oxford*.
2023. The microstructure of AMMs. *Seminar for master students, University of Edinburgh*.
2023. Optimal Liquidity Provision in AMMs. *KCL Financial Mathematics seminar, King's College London*.

Contributed

2026. Price formation in memory pools. *NBER Summer Institute, Cambridge, MA*.
2026. Price formation in memory pools. *6th Annual CBER Conference, New York*.
2026. Price formation in memory pools. *EEA-ESEM (Econometric Society), Dublin*.
2026. Liquid staking and the limits of policy. *Designing DeFi Conference, Columbia Business School*.
2025. Liquid staking and the limits of policy. *CBER Crafting the Cryptoeconomy Conference, Columbia University*.

2025. Price formation in memory pools. *Warwick Business School PhD Workshop on Financial Technology*.
2024. Price formation in memory pools. *CBER Cryptoeconomy Conf. Center for Digital Finance, Columbia University*.
2024. Mathematics of DeFi. *Workshop on Mathematical Insights of Markets, Control, and Learning. Ecole Polytechnique*.
2024. Advances in automated market making. *Oxford-Princeton Workshop XIII, Princeton University*.
2024. Equilibrium liquidity in Decentralised Markets. *Bachelier Finance Society – World Congress*.
2023. Predictable Losses of Liquidity Provision in AMMs. *EURO Working Group, Rome*.
2023. AMMs beyond constant functions. *WBS Gillmore Centre Conference: DeFi & Digital Currencies, Warwick University*.
2023. Bandits for algorithmic trading. *EURO Working Group for Commodities and Financial Modelling, Abu-Dhabi*.
2023. Bandits in algorithmic trading. *Fields-CFI Conf. on Advances in Math Finance and Insurance, University of Toronto*.
2023. Optimal liquidity Provision in AMMs. *SIAM Conference on Financial Mathematics, Philadelphia*.
2023. Optimal liquidity provision and execution in AMMs. *Oxford-ETH Seminar, University of Oxford*.